

Graph Statistics

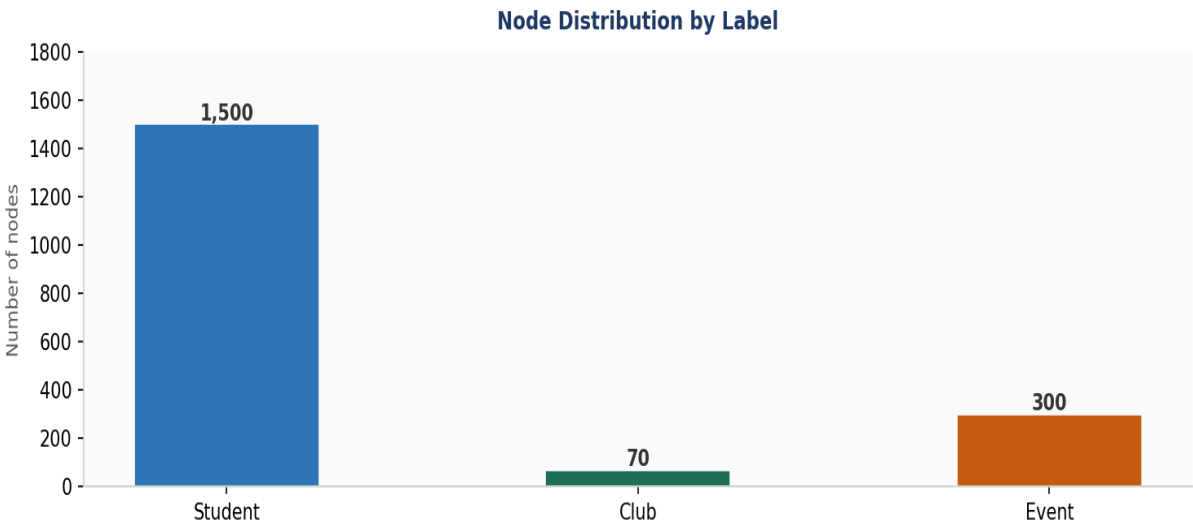
Student Social Network — Neo4j

1. Overview

This document presents the main structural statistics of the Student Social Network graph stored in Neo4j. The graph contains three node labels — Student, Club, Event — and three relationship types — EST_AMI_DE, MEMBRE_DE, A_PARTICIPE_A.

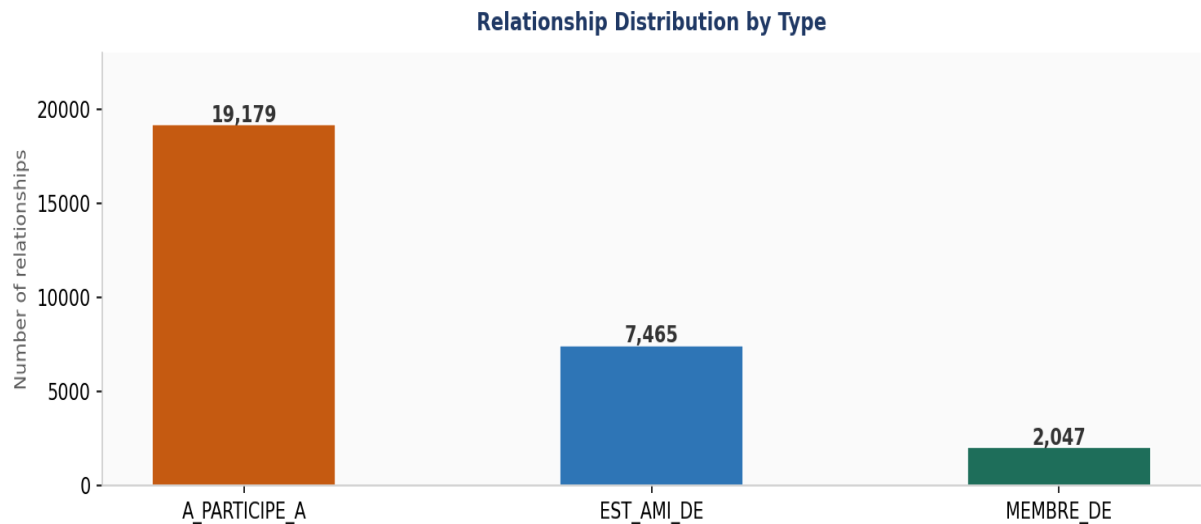
2. Node Statistics

Label	Number of Nodes
Student	1,500
Club	70
Event	300
Total	1,870



3. Relationship Statistics

Relationship Type	Number of Relationships
A_PARTICIPE_A	19,179
EST_AMI_DE	7,465
MEMBRE_DE	2,047
Total	28,691



4. Key Metrics

0.0066 Graph Density (friendship network)	9.95 Average Degree (friends per student)
0.127 Global Clustering Coefficient	28,691 Total Relationships

Graph Density

The density of the student friendship network is **0.0066**. This measures the proportion of existing friendship connections compared with the maximum possible number of friendship connections. A value close to 0 indicates a sparse network, which is consistent with real-world social networks where each person knows only a small fraction of all others.

Average Degree

The average degree of the student friendship network is **9.95**. This means that each student has approximately 9 to 10 friendship connections on average.

Global Clustering Coefficient

The global clustering coefficient is **0.127**. This measures the tendency of students who share a common friend to also be connected to each other. A value of 0.127 indicates a moderate level of local clustering, meaning that roughly 1 in 8 pairs of a student's friends are also friends with each other.

5. Summary

Statistic	Value
Total Nodes	1,870
Total Relationships	28,691
Students	1,500
Clubs	70
Events	300
EST_AMI_DE	7,465
MEMBRE_DE	2,047
A_PARTICIPE_A	19,179
Graph Density	0.0066
Average Degree	9.95
Global Clustering Coefficient	0.127